Toward Reliable Consensus in Multi-Agent LLM Systems

Research Direction Exploration

Your Name

Graduate Research Meeting

October 9, 2026

Research Motivation

Distributed Systems $\times$ Multi-Agent LLMs

Research Question

How can multiple LLM agents make reliable collective decisions?

Papers Studied

1. **The Byzantine Generals Problem**
   Lamport et al., 1982

2. **Tight Bound on Mobile Byzantine Agreement**
   Bonnet et al., DISC 2014

3. **Robust Multi-Agent LLMs under Byzantine Faults**
   Recent work under investigation

Classical Byzantine Agreement

Agreement

Validity

Termination

Mobile Byzantine Agreement



$$\boxed{n \geq 5t + 1}$$

- Mobile Byzantine faults
- Corrupted internal states
- Agreement maintenance

A Challenge in Multi-Agent LLMs

$$[A, A, A, B, B]$$

$$\text{Majority} = A$$

$$\text{Correct Answer} = B$$

Key Observation

Agreement does not imply answer correctness.

Current Research Direction

LLM Proposals

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Deterministic Verification

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Fault-Tolerant Agreement

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Verified Decision

Open Questions and Next Steps

Open Questions

- How should validity be defined?
- How can verification cost be reduced?
- How can safety be guaranteed under faults?

Next Steps

- Study validated Byzantine agreement
- Select a verifiable task
- Build a small prototype

References



L. Lamport, R. Shostak, and M. Pease.
The Byzantine Generals Problem.
ACM TOPLAS, 1982.



F. Bonnet, X. Défago, T. D. Nguyen, and M. Potop-Butucaru.
Tight Bound on Mobile Byzantine Agreement.
DISC, 2014.



M. Castro and B. Liskov.
Practical Byzantine Fault Tolerance.
OSDI, 1999.